

ANSWER KEY OF HOMEWORK 1

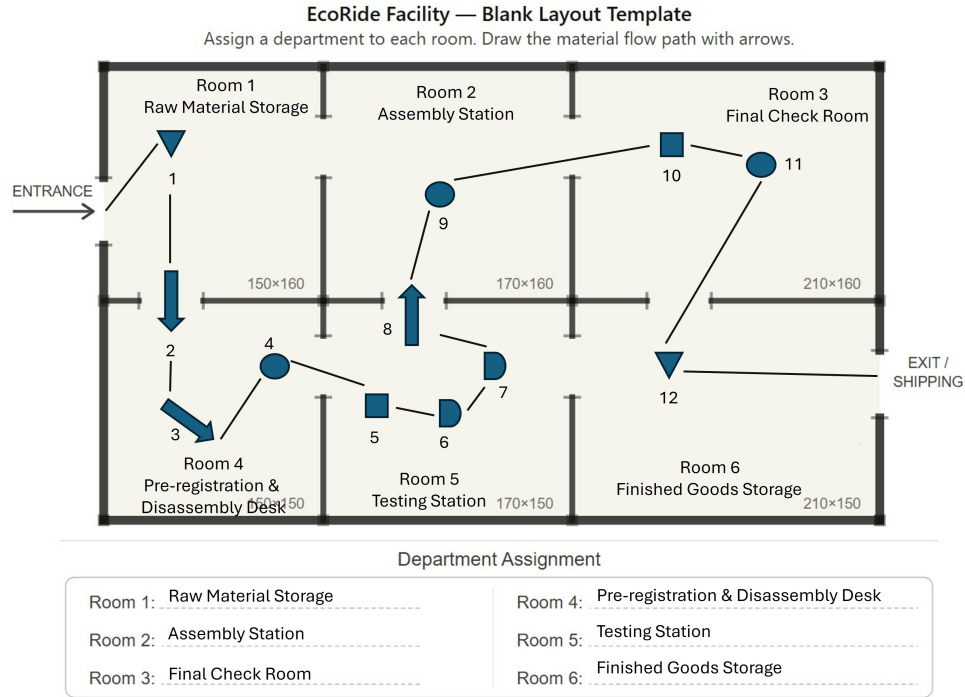
1. EcoRide Scooter Refurbishment Facility:

a) Ordered Steps with Flow Process Chart Symbols

Order	Item	Description	Symbol Name	Symbol
1	(v)	Used scooters arriving from outside are held in the main warehouse under authorization until they are ready to be processed.	Storage	▽
2	(x)	A scooter is taken from the warehouse and transported to the pre-registration desk by a cart.	Transportation	→
3	(iii)	After the intake processing steps are completed at the desk, the registered scooter is transferred to the disassembly station via a conveyor belt.	Transportation	→
4	(vii)	The battery and motor are separated from the frame.	Operation	○
5	(xii)	The battery and motor are sent to the testing station, where they are connected to measurement devices and checked against quality standards.	Inspection	□
6	(i)	The scooter frame remains stationary in front of the station until the next assembly worker becomes available.	Delay	▷
7	(viii)	Once the test outcome is recorded, the battery and motor are placed on hold alongside the already-queued frame until a worker becomes available.	Delay	▷
8	(xi)	The frame and the battery/motor set are transported to the relevant assembly table.	Transportation	→
9	(iv)	The refurbished battery and motor are mounted onto the frame.	Operation	○
10	(vi)	The fully assembled scooter is examined in the test room according to riding and safety standards.	Inspection	□
11	(ii)	Scooters that have completed and passed all tests are boxed.	Operation	○
12	(ix)	Finished scooters are held in the storage area until they are picked up for shipment.	Storage	▽

Summary: 3 Operations (○), 3 Transportations (→), 2 Inspections (□), 2 Delays (▷), 2 Storages (▽) = 12 steps ✓ **Key reasoning for non-obvious steps:**

b) and c)



2. EcoRide Scooter Refurbishment Facility (cont.):

a) Part (a) — Capacity Utilization Rate and Efficiency

Given: Design Capacity = 150, Effective Capacity = 120, Actual Output = 96 scooters/day.

$$\text{Capacity Utilization Rate} = \frac{\text{Actual Output}}{\text{Design Capacity}} \times 100 = \frac{96}{150} \times 100 = \boxed{64\%}$$

$$\text{Efficiency} = \frac{\text{Actual Output}}{\text{Effective Capacity}} \times 100 = \frac{96}{120} \times 100 = \boxed{80\%}$$

The facility uses only 64% of its theoretical maximum but operates at 80% of what is realistically achievable under normal conditions.

b) i) Utilization Rates Under Normal Conditions

Given: $\lambda = 40$ scooters/day, $c_1 = 3$, $\mu_1 = 18$, $c_2 = 4$, $\mu_2 = 14$, rework rate = 10%.

Station 1 workload: All incoming orders go to Station 1:

$$\lambda_1 = 40 \text{ scooters/day}$$

Station 2 workload: Station 2 receives all scooters from Station 1 plus 10% of its own output as rework:

$$\lambda_2 = 40 + 0.10 \times \lambda_2 \implies 0.90 \lambda_2 = 40 \implies \lambda_2 = \frac{40}{0.90} \approx 44.44 \text{ scooters/day}$$

Utilization rates:

$$\rho_1 = \frac{\lambda_1}{c_1 \times \mu_1} = \frac{40}{3 \times 18} = \frac{40}{54} \approx \boxed{74.1\%}$$

$$\rho_2 = \frac{\lambda_2}{c_2 \times \mu_2} = \frac{44.44}{4 \times 14} = \frac{44.44}{56} \approx \boxed{79.4\%}$$

Both stations are below 100% — the system is stable under normal conditions.

ii) Peak Season

Given: $\lambda = 65$ scooters/day, $c_1 = 4$ (one worker added), $\mu_1 = 18$, $c_2 = 4$, $\mu_2 = 14$, rework rate = 20%.

Station 1 workload: $\lambda_1 = 65$

Station 2 workload:

$$\lambda_2 = 65 + 0.20 \times \lambda_2 \implies 0.80 \lambda_2 = 65 \implies \lambda_2 = \frac{65}{0.80} = 81.25 \text{ scooters/day}$$

New utilization rates:

$$\rho_1 = \frac{65}{4 \times 18} = \frac{65}{72} \approx \boxed{90.3\%}$$

$$\rho_2 = \frac{81.25}{4 \times 14} = \frac{81.25}{56} \approx \boxed{145.1\%}$$

Station 2 is severely overloaded ($\rho_2 > 100\%$) — **Station 2 is the bottleneck.**

Maximum system throughput: Station 2 is the binding constraint. Its maximum total processing rate is:

$$\lambda_2^{\max} = c_2 \times \mu_2 = 4 \times 14 = 56 \text{ scooters/day (including rework)}$$

Of these, 20% are rework and do not exit as valid outputs:

$$\text{Maximum Throughput} = 56 \times (1 - 0.20) = 56 \times 0.80 = \boxed{44.8 \text{ approved scooters/day}}$$

c) **Layout Type for Mass Production**

Answer: Product Layout (Assembly Line)

When EcoRide switches to mass-producing a single standardized model at very high volumes, a **product layout** is the most appropriate choice. In a product layout, workstations are arranged in the exact sequence of production steps, forming a dedicated assembly line. This suits the scenario perfectly: product variety is minimal (one model), and production volume is very high. Work is divided into standardized, repetitive tasks, equipment can be specialized, material handling costs are minimized, and unit costs drop significantly through economies of scale. This is in direct contrast to the current refurbishment process, which requires a flexible layout to handle each scooter's unique condition.

d) i) (- pts) **Effective Operating Time**

$$OT = 480 \times (1 - 0.10) \times 0.80 = 480 \times 0.90 \times 0.80 = \boxed{345.6 \text{ min/day}}$$

ii) (- pts) **Desired Cycle Time**

$$CT = \frac{OT}{D} = \frac{345.6}{240} = \boxed{1.44 \text{ min/unit}}$$

iii) **Completed ALB Table** Using the formulas:

$$\text{Theoretical \#} = \frac{\text{Time Std.}}{CT} \quad \text{Avg. Time} = \frac{\text{Time Std.}}{\text{Round Up}} \quad \% \text{ Loaded} = \frac{\text{Avg. Time}}{CT}$$

Task	Time Std.	Theoretical #	Round Up	Avg. Time	% Loaded
A	2.2	$2.2/1.44 = 1.528$	2	$2.2/2 = 1.100$	$1.100/1.44 = 76.4\%$
B	4.0	$4.0/1.44 = 2.778$	3	$4.0/3 = 1.333$	$1.333/1.44 = 92.6\%$ bottl.
C	1.5	$1.5/1.44 = 1.042$	2	$1.5/2 = 0.750$	$0.750/1.44 = 52.1\%$
D	2.9	$2.9/1.44 = 2.014$	3	$2.9/3 = 0.967$	$0.967/1.44 = 67.1\%$
E	3.7	$3.7/1.44 = 2.569$	3	$3.7/3 = 1.233$	$1.233/1.44 = 85.6\%$

Bottleneck: Task B — highest % loaded at 92.6%.

iv) **Adding One Station to the Bottleneck**

Task B goes from 3 to 4 stations:

$$\text{Avg. Time}_B = \frac{4.0}{4} = 1.000 \text{ min} \implies \% \text{ Loaded}_B = \frac{1.000}{1.44} = 69.4\%$$

After this addition, the new bottleneck is **Task E** at **85.6%**, which now has the highest % loaded among all tasks.

- v) **Combining Tasks A and C Separately:** A requires 2 stations, C requires 2 stations $\rightarrow$ **4 stations total.**

Combined (A + C):

$$\text{Combined time} = 2.2 + 1.5 = 3.7 \text{ min/unit}$$

$$\text{Theoretical \#} = \frac{3.7}{1.44} = 2.569 \implies \text{Round up to } \mathbf{3} \text{ stations}$$

$$\text{Avg. Time} = \frac{3.7}{3} = 1.233 \text{ min} \implies \% \text{ Loaded} = \frac{1.233}{1.44} = \mathbf{85.6\%}$$

Combining A and C **saves 1 station** (3 instead of 4) while increasing utilization from an average of 64.3% to 85.6% — improving overall efficiency and reducing labor cost.

3. A University Cafeteria:

- a) The rate parameter:

$$\lambda = \frac{1}{\mu} = \frac{1}{40} = 0.025$$

- b) The critical ratio formula and calculation:

$$F(x^*) = \frac{p - c}{p - s} = \frac{150 - 60}{150 - 20} = \frac{90}{130} = \frac{9}{13} \approx 0.692$$

where $p = 150$ TL (selling price), $c = 60$ TL (purchasing cost), and $s = 20$ TL (clearance price).

- c) Using the inverse of the CDF:

$$F(x^*) = 1 - e^{-\lambda x^*} = CR$$

$$1 - e^{-\lambda x^*} = 0.692$$

$$e^{-\lambda x^*} = 0.308$$

$$\ln(e^{-\lambda x^*}) = \ln(0.308)$$

$$-\lambda x^* = 1.1744$$

$$x^* = \frac{1.1744}{0.025} \approx \mathbf{47 \text{ sushi boxes}}$$

- d) New critical ratio (with $c = 40$ TL):

$$F(x^*) = \frac{p - c}{p - s} = \frac{150 - 40}{150 - 20} = \frac{110}{130} = \frac{11}{13} \approx 0.846$$

New optimal quantity:

$$1 - e^{-\lambda x^*} = 0.846$$

$$e^{-\lambda x^*} = 0.154$$

$$\ln(e^{-\lambda x^*}) = \ln(0.154)$$

$$-\lambda x^* = 1.8695$$

$$x^* = \frac{1.8695}{0.025} \approx \mathbf{75 \text{ sushi boxes}}$$

The optimal order quantity **INCREASES** from 47 to 75 boxes. This is because a lower purchasing cost c reduces the cost of overstocking. When c decreases, the numerator $(p - c)$ increases, making the critical ratio larger, which corresponds to a higher optimal order quantity.

4. EV Charging Stations:

Let the eight districts be labeled as follows: 1 = Beyoğlu, 2 = Şişli, 3 = Beşiktaş, 4 = Bakırköy, 5 = Küçükçekmece, 6 = Bağcılar, 7 = Güngören, 8 = Zeytinburnu.

Site	Districts Covered	Cost (TL)
S1 (Taksim Transit Hub)	{1, 2, 4, 5}	120,000
S2 (Mecidiyeköy Depot)	{2, 3, 6}	80,000
S3 (Etiler Road Junction)	{3, 6, 7}	70,000
S4 (Ataköy Coastal Complex)	{4, 5, 8}	100,000
S5 (Şişhane Interchange)	{1, 4, 8}	90,000
S6 (Bahçelievler Logistics Center)	{5, 7, 8}	85,000

The efficiency of each site is computed as:

$$\text{Efficiency}(S_i) = \frac{|\text{uncovered districts in } S_i|}{c_i}$$

a) Step-by-Step Greedy Algorithm

Iteration 1:

(i) $U' = \{1, 2, 3, 4, 5, 6, 7, 8\}$

	Site	Uncovered districts covered	Count	Cost	Efficiency
	S1	{1,2,4,5}	4	120,000	$4/120,000 \approx 0.0000\bar{3}$
	S2	{2,3,6}	3	80,000	$3/80,000 = 0.0000375$
(ii)	S3	{3,6,7}	3	70,000	$3/70,000 \approx 0.0000429$
	S4	{4,5,8}	3	100,000	$3/100,000 = 0.0000300$
	S5	{1,4,8}	3	90,000	$3/90,000 = 0.0000\bar{3}$
	S6	{5,7,8}	3	85,000	$3/85,000 \approx 0.0000353$

(iii) **Selected: S3** — highest efficiency ≈ 0.0000429 .

(iv) Newly covered: {3, 6, 7}. Covered so far: {3, 6, 7}.

Iteration 2:

(i) $U' = \{1, 2, 4, 5, 8\}$

	Site	Uncovered districts covered	Count	Cost	Efficiency
	S1	{1,2,4,5}	4	120,000	$4/120,000 \approx 0.0000\bar{3}$
	S2	{2}	1	80,000	$1/80,000 = 0.0000125$
(ii)	S4	{4,5,8}	3	100,000	$3/100,000 = 0.0000300$
	S5	{1,4,8}	3	90,000	$3/90,000 = 0.0000\bar{3}$
	S6	{5,8}	2	85,000	$2/85,000 \approx 0.0000235$

(iii) S1 and S5 are tied at efficiency $= 1/30,000 \approx 0.0000\bar{3}$. **S1 is selected** as it covers more uncovered districts (4 vs. 3).

(iv) Newly covered: {1, 2, 4, 5}. Covered so far: {1, 2, 3, 4, 5, 6, 7}.

Iteration 3:

(i) $U' = \{8\}$

	Site	Uncovered districts covered	Count	Cost	Efficiency
	S4	{8}	1	100,000	$1/100,000 = 0.0000100$
(ii)	S5	{8}	1	90,000	$1/90,000 \approx 0.0000111$
	S6	{8}	1	85,000	$1/85,000 \approx 0.0000118$

(iii) **Selected: S6** — highest efficiency ≈ 0.0000118 .

(iv) Newly covered: {8}. Covered so far: {1, 2, 3, 4, 5, 6, 7, 8}.

$U' = \emptyset$ — all districts are covered. **Algorithm terminates.**

b) Final Solution and Total Cost

Selected Site	Districts Served	Cost (TL)
S3 — Etiler Road Junction	Beşiktaş, Bağcılar, Güngören	70,000
S1 — Taksim Transit Hub	Beyoğlu, Şişli, Bakırköy, Küçükçekmece	120,000
S6 — Bahçelievler Logistics Center	Küçükçekmece, Güngören, Zeytinburnu	85,000
Total		275,000

$$\text{Total Installation Cost} = 70,000 + 120,000 + 85,000 = \boxed{275,000 \text{ TL}}$$

All 8 districts are covered by at least one charging station. ✓

5. Mars Rover Instrument Selection:

Recursive Relation (from lecture):

$$V(i, w) = \begin{cases} V(i-1, w) & \text{if } w_i > w \\ \max\{V(i-1, w), p_i + V(i-1, w - w_i)\} & \text{if } w_i \leq w \end{cases}$$

where $V(i, w)$ = maximum value using first i items with capacity w .

a) **Row 1 — Item A** ($w_A = 2, p_A = 30$):

- $w < 2$: $w_A > w \Rightarrow$ copy from above $\Rightarrow V(1, 0) = 0, V(1, 1) = 0$
- $w \geq 2$: $\max\{V(0, w), 30 + V(0, w - 2)\} = \max\{0, 30\} = 30$

Row 2 — Item B ($w_B = 3, p_B = 50$):

- $w < 3$: copy $\Rightarrow V(2, 0) = 0, V(2, 1) = 0, V(2, 2) = 30$
- $w \geq 3$: $\max\{V(1, w), 50 + V(1, w - 3)\}$

Row 3 — Item C ($w_C = 4, p_C = 60$):

- $w < 4$: copy from Row 2
- $w \geq 4$: $\max\{V(2, w), 60 + V(2, w - 4)\}$

Row 4 — Item D ($w_D = 5, p_D = 90$):

- $w < 5$: copy from Row 3
- $w \geq 5$: $\max\{V(3, w), 90 + V(3, w - 5)\}$

DP Table to Complete:

Capacity w	0	1	2	3	4	5	6	7	8
No instrument	0	0	0	0	0	0	0	0	0
A ($w = 2, p = 30$)	0	0	30	30	30	30	30	30	30
B ($w = 3, p = 50$)	0	0	30	50	50	80	80	80	80
C ($w = 4, p = 60$)	0	0	30	50	60	80	90	110	110
D ($w = 5, p = 90$)	0	0	30	50	60	90	90	120	140

Explicit calculations for required cells:

$V(2, 4)$ — considering Item B ($w_B = 3 \leq 4$):

$$V(2, 4) = \max\{V(1, 4), 50 + V(1, 4 - 3)\} = \max\{30, 50 + V(1, 1)\} = \max\{30, 50 + 0\} = \boxed{50}$$

$V(3, 7)$ — considering Item C ($w_C = 4 \leq 7$):

$$V(3, 7) = \max\{V(2, 7), 60 + V(2, 7 - 4)\} = \max\{80, 60 + V(2, 3)\} = \max\{80, 60 + 50\} = \boxed{110}$$

b) **Maximum Scientific Value:**

$$V(4, 8) = \max\{V(3, 8), 90 + V(3, 8 - 5)\} = \max\{110, 90 + V(3, 3)\} = \max\{110, 90 + 50\} = \boxed{140 \text{ points}}$$

c) **Traceback: Which Instruments to Select**

Starting from $V(4, 8) = 140$:

- **Step 1:** $V(4, 8) = 140$. Since $90 + V(3, 3) = 140 > V(3, 8) = 110 \Rightarrow$ **Item D is selected.**
Remaining capacity = $8 - 5 = 3$.
- **Step 2:** $V(3, 3) = 50$. Since $w_C = 4 > 3 \Rightarrow$ Item C cannot fit $\Rightarrow$ **Item C is NOT selected.**
Move to $V(2, 3) = 50$.
- **Step 3:** $V(2, 3) = 50$. Since $50 + V(1, 0) = 50 > V(1, 3) = 30 \Rightarrow$ **Item B is selected.**
Remaining capacity = $3 - 3 = 0$.
- **Step 4:** Remaining capacity = $0 \Rightarrow$ **Item A is NOT selected.**

Optimal selection: Instruments B and D

Instrument	Weight	Value
B (Ground-Penetrating Radar)	3 kg	50 pts
D (Geochemistry Analyzer)	5 kg	90 pts
Total	8 kg ✓	140 pts ✓